8 When a drawing pin is dropped it can land point down or point up.

Lucy, Mel and Tom each dropped the drawing pin a number of times.

The table shows the number of times the drawing pin landed point down and the number of times the drawing pin landed point up for each person.

	Lucy	**Mel**	**Tom**
point down	31	53	16
point up	14	27	9

Rachael is going to drop the drawing pin once.

(a) Whose results will give the best estimate for the probability that the drawing pin will land point up?
Give a reason for your answer.

Mel: 53 + 27 = 80 throws, the largest number of trials.

Lucy made 45 throws and Tom made 25.

1 mark
Larger samples give more reliable probability estimates

(1)

Stuart is going to drop the drawing pin twice.

(b) Use all the results in the table to work out an estimate for the probability that the drawing pin will land point up the first time and point down the second time.

Total up = 14 + 27 + 9 = 50

Total down = 31 + 53 + 16 = 100; total throws = 150

Treat the two drops as independent.

$$P(\text{up then down}) = \frac{50}{150} \times \frac{100}{150} = \frac{2}{9}$$

1 mark
Use all results and multiply the two probabilities

1 mark
Correct probability

$\frac{2}{9}$ **(2)**

(Total for Question 8 is 3 marks)